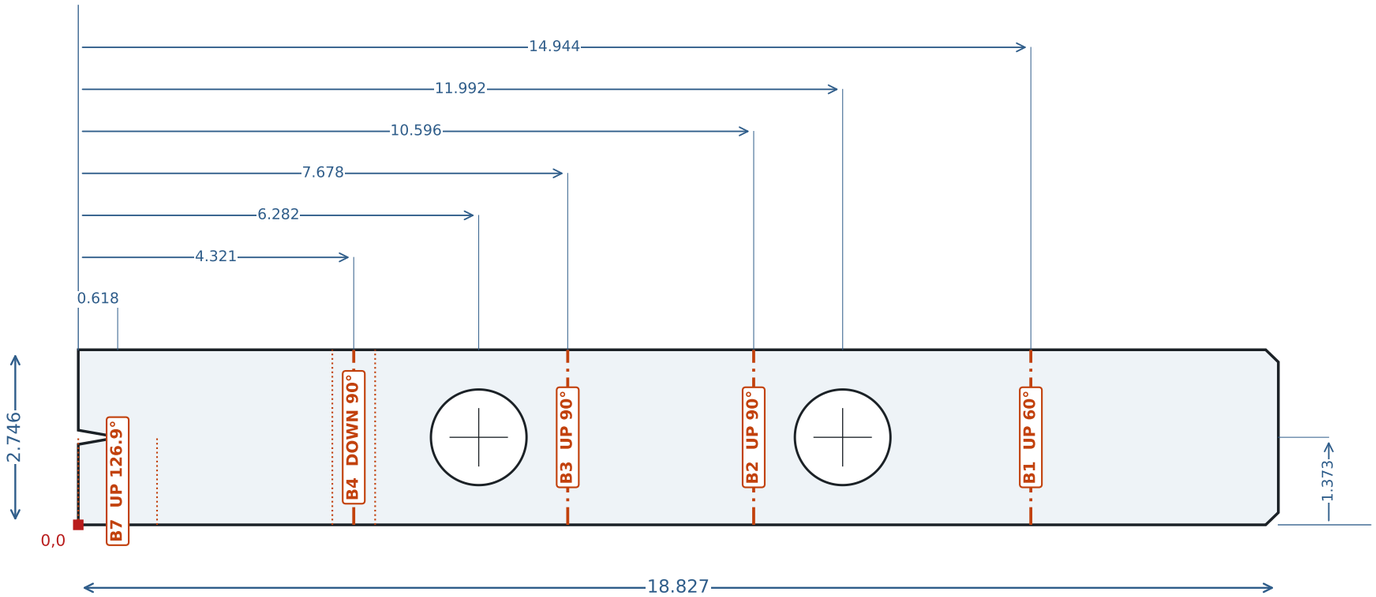
FR-4-1 · Wrap strip (ear, top, front, bottom, slant)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 18.827″ × 2.746″ · ≈ 1.63 lb each · QTY 1 FR + 1 FL = 2 per vehicle

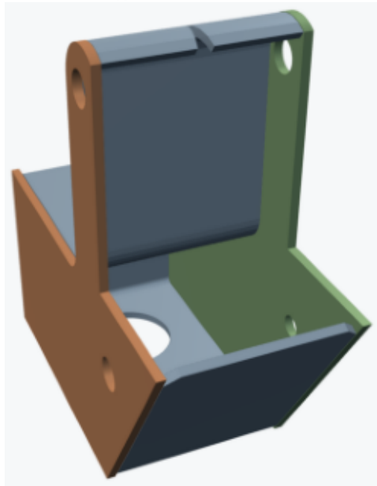
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-4-1 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	60.0°	UP (toward)	R=t 0.120	5	X = 14.944
B2	90.0°	UP (toward)	R=t 0.120	4	X = 10.596
B3	90.0°	UP (toward)	R=t 0.120	3	X = 7.678
B4	90.0°	DOWN (away)	R 0.375	2	X = 4.321
		radius zone	edges		X = 3.986 / 4.657
B7	126.9°	UP (toward)	R 0.505	1	X = 0.618
		radius zone	edges		X = 0.000 / 1.236

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.500	6.282	1.373	laser
H2	Ø1.500	11.992	1.373	laser

FILES

- FR-4-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-4-1_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat.
   Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* arc C-D bent over R 0.505 (outer R 0.625 = side plate arc); R 0.375 at E (concave, opposite way); no corner reliefs; strip width = 3" - 2t - 0.015

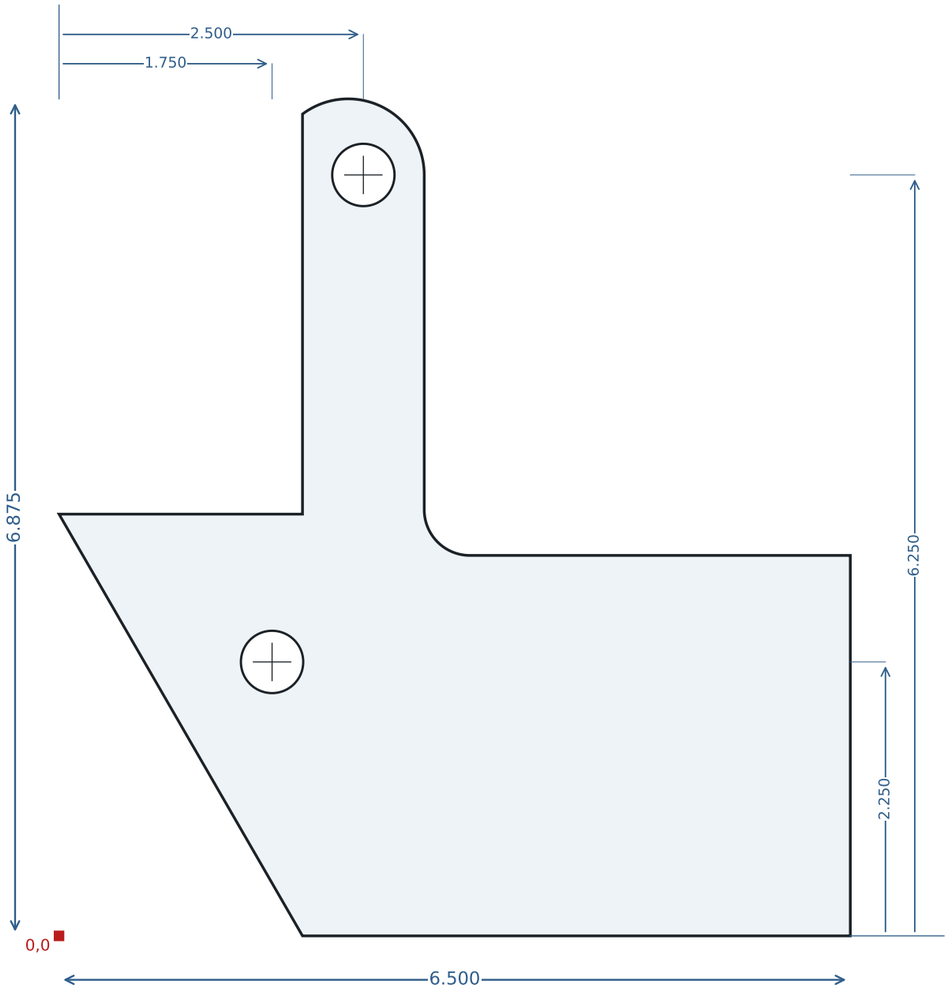
FR-4 Bracket: wrap strip + 2 side plates + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-4-1 · Wrap strip (ear, top, front, bottom, slant)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/4**

FR-4-2 · Side plate A (2 holes Ø0.512)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 6.500″ × 6.875″ · ≈ 0.71 lb each · QTY 1 FR + 1 FL = 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-4-2 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)

flat part - no bends

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.512	1.750	2.250	laser
H2	Ø0.512	2.500	6.250	laser

FILES

- FR-4-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-4-2_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* flat, no bends

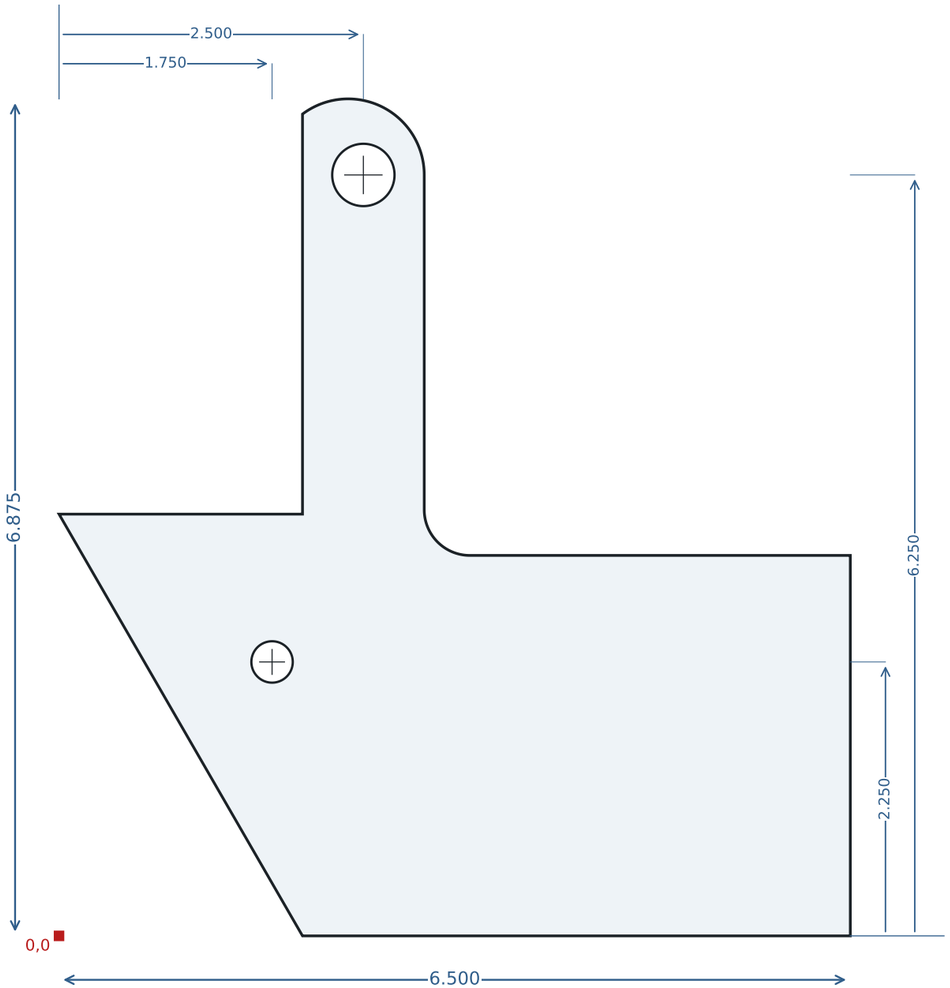
FR-4 Bracket: wrap strip + 2 side plates + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-4-2 · Side plate A (2 holes Ø0.512)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 2/4**

FR-4-3 · Side plate B (holes Ø0.512 + Ø0.34)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 6.500″ × 6.875″ · ≈ 0.71 lb each · QTY 1 FR + 1 FL = 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-4-3 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)

flat part - no bends

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.340	1.750	2.250	laser
H2	Ø0.512	2.500	6.250	laser

FILES

- FR-4-3_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-3_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-4-3_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* flat, no bends

FR-4 Bracket: wrap strip + 2 side plates + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-4-3 · Side plate B (holes Ø0.512 + Ø0.34)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 3/4**

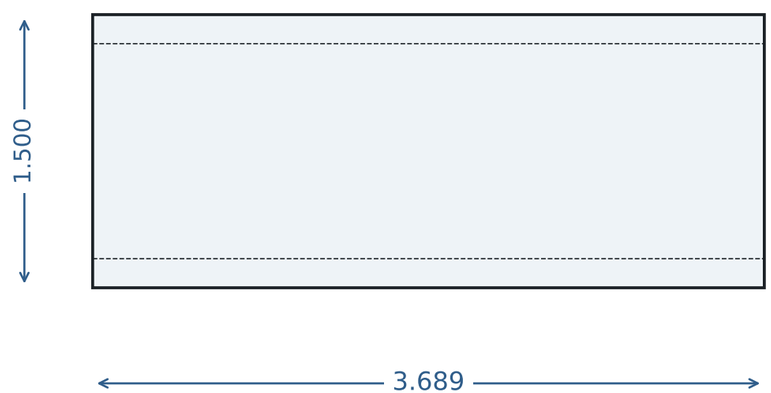
FR-4 · tube cut list

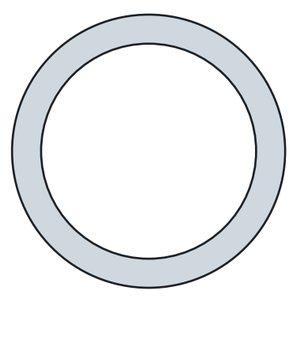
MATERIAL: DOM round steel tube A513

WALL THICKNESS: OD 1.500 × ID 1.181 (wall 0.160″)

Tube OD 1.5 / ID 1.181 cut 3.689″ (through bottom and top)

Stock: DOM round steel tube OD 1.500 × ID 1.181 (wall 0.160″)





section, wall 0.1595″

Cut length 3.689″ ±1/32″, square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

FR-4 Bracket: wrap strip + 2 side plates + tube		**DOM round steel tube A513**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
Tube cut list		**QTY 1 per side × 2 (FR + FL) = 2 per vehicle**	Tol.: cut ±1/32″	2026-10-31	**SHEET 4/4**